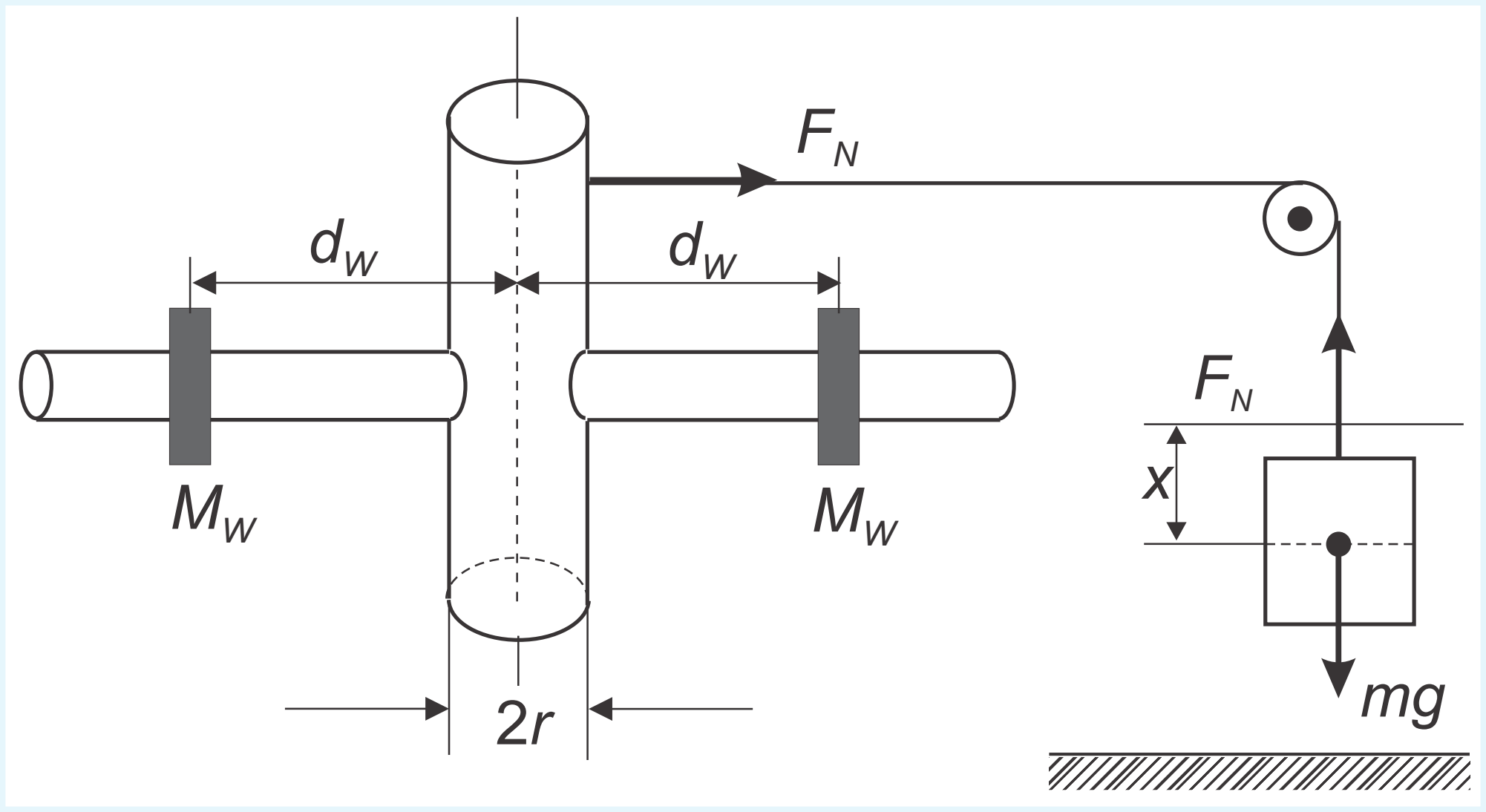
F_N
d_w
d_w
M_w
M_w
F_N
x
mg
$2r$